ARITHMETIC PROGRESSIONS

Chapter
5

Exercise-1

Part - 1 (Basic Multiple Choice Questions)

1. Which of the following lists of numbers forms an Arithmetic Progression? **[NCERT]**
(a) –5, –1, 3, 7, . . . (b) 2, 4, 8, 16, . . . (c) 1, 4, 9, 16, . . . (d) 3, 3.5, 4.5, 6, . . .

2. The common difference of the AP $\frac{1}{2},-\frac{5}{2},-\frac{11}{2},\ldots\ldots$ is **[Mod. NCERT]**
(a) 3 (b) –3 (c) –2 (d) $-\frac{5}{2}$

3. For the AP 2, 7, 12, . . . , the 10^{th} term is **[NCERT]**
(a) 45 (b) 47 (c) 52 (d) 42

4. If the n^{th} term of an AP is $a^n = 3n + 4$, then its common difference is **[Mod. NCERT Exemplar]**
(a) 3 (b) 4 (c) 7 (d) 1

5. The 6^{th} term of the AP whose first term is 2 and common difference is 3 is **[NCERT]**
(a) 17 (b) 20 (c) 15 (d) 14

6. The sum of the first *n* terms of an AP with first term *a* and last term *l* is **[NCERT]**
(a) $\frac{n}{2}(a-l)$ (b) $\frac{n}{2}(a+l)$ (c) $n(a+l)$ (d) $\frac{n}{2}l$

7. How many terms are there in the AP 7, 13, 19, . . . , 205? **[Mod. NCERT]**
(a) 33 (b) 34 (c) 35 (d) 32

8. If $a_3 = 5$ and $a_7 = 13$ for an AP, then a_{11} equals **[Competency Based]**
(a) 25 (b) 21 (c) 17 (d) 29

9. If $2x$, $x + 10$, $3x + 2$ are three consecutive terms of an AP, then x = **[Mod. NCERT Exemplar]**
(a) 6 (b) 8 (c) 10 (d) 18

10. The sum of the first 20 positive integers is **[NCERT]**
(a) 210 (b) 200 (c) 190 (d) 420

11. If the first term of an AP is 4 and the common difference is –2, then S_5 equals **[Mod. NCERT]**
(a) 0 (b) 10 (c) –10 (d) 20

12. In an AP, if $a_{18} - a_{14} = 16$, then the common difference is **[Competency Based]**
(a) 4 (b) 2 (c) 8 (d) – 4

Part - 2 (Basic Subjective Type Questions)

1. Find the common difference of the AP **[NCERT]**

   $\sqrt{2}, \sqrt{2}+\frac{3}{2}, \sqrt{2}+3, ...$ and write its next two terms.

2. Find the 18th term of the AP **[Mod. NCERT]**

   $\frac{5}{2}, 2, \frac{3}{2}, 1, ...$

3. Which term of the AP **[Mod. NCERT]**

   $\sqrt{3}, \sqrt{3}+2, \sqrt{3}+4, ...$ is $\sqrt{3}+48$?

4. Find the sum of the first 20 terms of the AP **[NCERT]**

   $\frac{1}{3}, \frac{5}{6}, \frac{4}{3}, \frac{11}{6}, ...$

5. The 4th term of an AP is 11 and the 8th term is 23. Find its first term and common difference. **[Mod. NCERT]**
6. Find the sum of all the numbers between 1 and 51 that leave a remainder of 2 when divided by 3. **[Competency Based]**
7. The sum of the first *n* terms of an AP is given by $S_n = \frac{n}{2}(n+5)$. Find the AP. **[Mod. NCERT Exemplar]**

Exercise-2

Part - 1 (Advanced Multiple Choice Questions)

1. For an AP, the sum of the first *m* terms equals the sum of the first *n* terms, where $m \neq n$. Then the sum of its first ($m + n$) terms is **[Competency Based]**

   (a) $m + n$ (b) 0 (c) mn (d) 1

2. A sum of Rs. 700 is to be given as seven cash prizes to students, each prize being Rs. 20 less than the one before it. The value of the highest prize is **[Mod. NCERT]**

   (a) Rs. 160 (b) Rs. 140 (c) Rs. 180 (d) Rs. 150

3. In an AP, the sum of the first p terms is q and the sum of the first q terms is $p \neq q$. The sum of its first (p + q) terms is **[Competency Based]**

   (a) p + q (b) – (p + q) (c) 0 (d) pq

4. How many terms of the AP 45, 39, 33, . . . must be taken so that their sum is 180? **[Competency Based]**

   (a) 6 only (b) 10 only (c) 6 or 10 (d) 5 or 12

5. The first negative term of the AP 17, $16\frac{1}{5}, 15\frac{2}{5}, ...$ is its **[Competency Based]**

   (a) 22nd term (b) 23rd term (c) 24th term (d) 25th term

6. In an AP, the m^{th} term is $\frac{1}{n}$ and the n^{th} term is $\frac{1}{m}$. Then its $(mn)^{th}$ term is **[NCERT Exemplar]**

(a) $\frac{1}{mn}$ (b) 1 (c) $\frac{m+n}{mn}$ (d) mn

7. Three numbers are in AP. Their sum is 24 and the sum of their squares is 200. The numbers are **[Competency Based]**

(a) 6, 8, 10 (b) 4, 8, 12 (c) 5, 8, 11 (d) 7, 8, 9

8. The ratio of the sums of the first n terms of two APs is (7n +1) : (4n +27). The ratio of their 11^{th} terms is **[NCERT Exemplar]**

(a) 4 : 3 (b) 7 : 4 (c) 3 : 4 (d) 11 : 10

9. If $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in AP, then which of the following are in AP? **[Competency Based]**

(a) a, b, c (b) a^2, b^2, c^2 (c) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ (d) a^3, b^3, c^3

10. If *a, b, c* are in AP, then $\frac{1}{\sqrt{b}+\sqrt{c}}, \frac{1}{\sqrt{c}+\sqrt{a}}, \frac{1}{\sqrt{a}+\sqrt{b}}$ are in **[Competency Based]**

(a) AP (b) GP (c) HP (d) None of these

11. The number of terms common to the two APs 3, 7, 11, . . . , 407 and 2, 9, 16, . . . , 709 is **[Competency Based]**

(a) 13 (b) 14 (c) 15 (d) 12

12. An AP consists of 37 terms. The sum of its three middle-most terms is 225 and the sum of its last three terms is 429. The first term of the AP is **[NCERT Exemplar]**

(a) 3 (b) 7 (c) 5 (d) 1

13. The sum of the first *n* terms of an AP is $Sn = 3n^2 + 5n$. Which term of this AP equals 164? **[Competency Based]**

(a) 26th (b) 27th (c) 28th (d) 25th

14. Four numbers are in AP. Their sum is 20 and the sum of their squares is 120. The smallest of the four numbers is **[Competency Based]**

(a) 2 (b) 3 (c) 4 (d) 1

15. A debt of Rs. 3600 is to be paid in 40 annual instalments that form an AP. When 30 instalments are paid, one-third of the debt remains unpaid. The first instalment is **[NCERT Exemplar]**

(a) Rs. 51 (b) Rs. 53 (c) Rs. 49 (d) Rs. 55

16. The sum of all two-digit numbers that leave remainder 1 when divided by 4 is **[Competency Based]**

(a) 1210 (b) 1200 (c) 1188 (d) 1320

17. An AP has 50 terms. The sum of its first 10 terms is 210 and the sum of its last 15 terms is 2565. Its first term is **[NCERT Exemplar]**

(a) 3 (b) 5 (c) 2 (d) 4

18. The houses in a row are numbered 1, 2, 3, . . . , 49. There is a house numbered *x* such that the sum of the house numbers preceding it equals the sum of those following it. The value of *x* is **[Competency Based]**

(a) 35 (b) 36 (c) 34 (d) 30

19. The angles of a quadrilateral are in AP and the greatest angle is double the smallest. The greatest angle is **[Competency Based]**

(a) 120º (b) 90º (c) 150º (d) 108º

20. In an AP, 7 times the 7th term is equal to 11 times the 11th term. Then its 18th term is **[Competency Based]**

(a) 7 (b) 11 (c) 0 (d) 18

21. If 1 + 4 + 7 + 10 +...........+ x = 287, then the value of x is **[Competency Based]**

(a) 40 (b) 43 (c) 37 (d) 46

22. If a, b, c, d, e are in AP, then the value of a – 4b + 6c – 4d + e is **[Competency Based]**

(a) 1 (b) 0 (c) 2(a + e) (d) 4c

23. If m arithmetic means are inserted between 1 and 31 so that the ratio of the 7th mean to the (m – 1)th mean is 5 : 9, then m = **[Competency Based]**

(a) 14 (b) 12 (c) 16 (d) 10

24. The sum of the first n terms of an AP is Sn = 4n.n^2. The 10th term of this AP is **[Competency Based]**

(a) – 15 (b) – 60 (c) – 13 (d) – 17

25. For what value of n is the nth term of the AP 3, 10, 17, . . . equal to the nth term of the AP 63, 65, 67, . . . ?

[Mod. NCERT]

(a) 13 (b) 12 (c) 14 (d) 11

26. Three consecutive terms of an AP have sum –3 and product 8. The terms are **[Competency Based]**

(a) – 4, –1, 2 (b) –2, –1, 0 (c) 1, –1, –3 (d) –5, –1, 3

Part - 2 (Advanced Subjective Type Questions)

1. For an AP, the sum of the first p terms equals the sum of the first q terms, where $p \neq q$. Prove that the sum of its first (p + q) terms is zero. **[NCERT Exemplar]**

2. Divide 32 into four parts which are in AP such that the ratio of the product of the first and the fourth to the product of the second and the third is 7 : 15. **[NCERT Exemplar]**

3. In an AP, the mth term is $\frac{1}{n}$ and the nth term is $\frac{1}{m}$. Prove that the sum of its first mn terms is

$\frac{1}{2}(mn+1)$. **[NCERT Exemplar]**

4. The sum of three numbers in AP is 12 and the sum of their cubes is 288. Find the numbers.

[Mod. NCERT Exemplar]

5. A manufacturer produced 600 television sets in the third year and 700 sets in the seventh year. Assuming production increases by a fixed number each year, find **[Mod. NCERT]**

(i) the production in the first year,

(ii) the production in the 10th year,

(iii) the total production in the first 7 years.

6. If the pth, qth and rth terms of an AP are a, b and c respectively, prove that a(q – r) + b(r – p) + c(p – q) = 0.

[NCERT Exemplar]

7. The first term of an AP is 5, its last term is 45 and the sum of all its terms is 400. Find the number of terms and the common difference. **[Mod. NCERT]**

8. How many terms of the AP 9, 17, 25, . . . must be taken so that their sum is 636? **[NCERT]**

9. The sum of the 4th and 8th terms of an AP is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the AP. **[NCERT Exemplar]**

10. A spiral is made of successive semicircles, with centres alternately at *A* and *B*, starting with centre *A*, of radii 0.5, 1.0, 1.5, 2.0, . . . cm. Find the total length of the spiral made up of thirteen consecutive semicircles.

$\left(\text{Take } \pi = \frac{22}{7}\right)$ **[NCERT]**

11. In a flower bed there are 23 rose plants in the first row, 21 in the second, 19 in the third, and so on. If there are 5 rose plants in the last row, how many rows are there, and how many rose plants in all? **[NCERT]**

12. Show that the sum of an AP whose first term is a, second term is b and last term is c is equal to $\frac{(a+c)(b+c-2a)}{2(b-a)}$.

[NCERT Exemplar]

13. If S_n denotes the sum of the first n terms of an AP, prove that $S_{30} = 3(S_{20} - S_{10})$. **[Competency Based]**

14. The digits of a three-digit positive number are in AP and their sum is 15. The number obtained by reversing the digits is 594 less than the original number. Find the original number. **[Competency Based]**

15. Find three numbers in AP whose sum is 15 and whose product is 80. **[Mod. NCERT]**

16. Two APs have the same common difference. The difference between their 100th terms is 100. What is the difference between their 1000th terms? Justify. **[NCERT]**

17. 150 workers were engaged to finish a piece of work in a certain number of days. Four workers dropped out on the second day, four more on the third day, and so on. It took 8 more days to finish the work than originally planned. Find the number of days in which the work was actually completed. **[NCERT Exemplar]**

18. Is it possible for an AP whose sum of the first n terms is $S_n = 3n^2 - 4n$ to have a term equal to 50? Justify, and if possible state which term. **[Competency Based]**

19. Can $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ be three consecutive terms of an AP? Justify your answer. **[Competency Based]**

Part - 3 (Advanced Miscellaneous Questions)

Assertion-Reason. In each question two statements are given – Assertion (A) and Reason (R).
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is *not* the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

1. **Assertion (A) :** A sequence whose n^{th} term is $a_n = An^2 + Bn + C$ with $A \neq 0$ is never an AP.
Reason (R) : The n^{th} term of an AP is always a polynomial of degree at most one in n. **[Competency Based]**

2. **Assertion (A) :** If $S_n = 3n^2 + 2n$ is the sum of the first n terms of an AP, then its common difference is 6.
Reason (R) : For any AP, the common difference equals the coefficient of n in S_n. **[Competency Based]**

3. **Assertion (A) :** 301 is a term of the AP 5, 11, 17, 23,...
Reason (R) : The n^{th} term of this AP is $a_n = 6n - 1$. **[Competency Based]**

4. **Assertion (A) :** For an AP with common difference d, the difference $a_{20} - a_{15} = 5d$.

Reason (R) : The sum of the first n terms of an AP with first term a and last term l is $S_n = \frac{n}{2}(a + l)$.

[Competency Based]

• **Match the Following.** Match each entry in Column I with the correct entry in Column II.

5 **Match 1.** **[Competency Based]**

Column I	Column II (common difference)
(A) $\frac{1}{3}, \frac{5}{12}, \frac{1}{2}, ...$	(i) -2
(B) $\sqrt{18}, \sqrt{50}, \sqrt{98}, ...$	(ii) 4
(C) AP with $a_n = 7 - 2n$	(iii) $\frac{1}{12}$
(D) AP with $S_n = 2n^2 - n$	(iv) $2\sqrt{2}$

6. **Match 2.** **[Competency Based]**

Column I	Column II (value)
(A) Number of terms in the AP 7, 10, 13,...,76	(i) 400
(B) 10^{th} term from the end of 4, 9, 14 ,..., 254	(ii) 19
(C) Sum of the first 20 odd natural numbers	(iii) 24
(D) Value of n(> 0) for which $S_n = 0$ in 27, 24, 21, ...	(iv) 209

True / False. State whether each statement is True or False, with justification.

7. If the n^{th} term of a sequence is $a_n = 2n^2 + 3$, then the sequence is an AP. **[Competency Based]**

8. Every arithmetic progression is also a geometric progression. **[Competency Based]**

9. If three numbers a, b, c are in AP, then $b = \dfrac{a+c}{2}$. **[Competency Based]**

10. The sum of the first n even natural numbers is n^2. **[Competency Based]**

11. If the common difference of an AP is negative, then all of its terms must be negative. **[Competency Based]**

Exercise-3

(Case Based Questions)

1. Ms. Sheela visited a store near her house and found that the glass jars are arranged above the other in a specific pattern. On the top layer there are 3 jars. In the next layer there are 6 jars. In the 3rd layer from the top there are 9 jars and so on till the 8th layer. **[CBSE SQP 2024]**

(a) Write an A.P whose terms represent the number of jars in different layers starting from top . Also, find the common difference.

(b) It is possible to arrange 34 jars in a layer if this pattern is continued? Justify your answer.

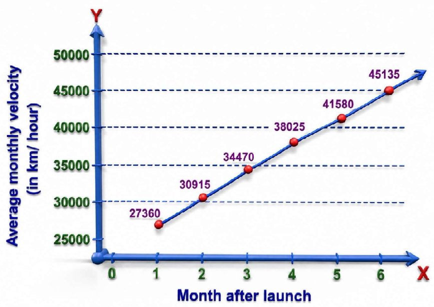
(c) If there are 'n' number of rows in a layer then find the expression for finding the total number of jars in terms of n. Hence Find S_8.

(d) The shopkeeper added 3 jars in each layer. How many jars are there in the 5th layer from the top?

[CBSE SQP 2024]

2. In space exploration missions, ion propulsion engine is an efficient way of travelling. The first such engine was used in Deep Space 1 spacecraft. It produced a constant acceleration. Given below are approximate velocities of the engine.

The initial average velocity of the engine in its first month was 27360 km/hour. When the spacecrafts passed the asteroid, Braille, it reached an average velocity of 55800 km/hour. Based on the first 6 months of Deep Space 1's monthly average velocity, the following table was created.



Month after launch	Average monthly velocity (in km/hour)
1	27360
2	30915
3	34470
4	38025
5	41580
6	45135

(a) Does the average monthly velocity (in km/hour) form an arithmetic progression? Justify your answer.

(b) The distance travelled by the spacecraft in the first 10 months (or 7300 hours) can be expressed as 7300p km where p is the sum of average monthly velocity for the first 10 months. Find p. Show your work.

(c) The spacecraft passed the comet, Borelly, 15 months after it passed Braille. Find the average monthly velocity of the spacecraft when it passed Borelly. Show your work.

(d) After how many months did the spacecraft pass Braille? Show your work.

[CBSE Question Bank 2023]

3. The school auditorium was to be constructed to accommodate at least 1500 people. The chairs are to be placed in concentric circular arrangement in such a way that each succeeding circular row has 10 seats more than the previous one.

(a) If the first circular row has 30 seats, how many seats will be there in the 10^{th} row?

(a) 120 (b) 110

(c) 100 (d) 130

(b) For 1500 seats in the auditorium, how many rows need to be there?

OR

If 1500 seats are to be arranged in the auditorium, how many seats are still left to be put after 10^{th} row?

(c) If there were 17 rows in the auditorium, how many seats will be there in the middle row?

(a) 100 (b) 130

(c) 110 (d) 120 **[Mod. CBSE SQP Std. 2022]**

4. Manpreet Kaur is the national record holder for women in the shot-put discipline. Her throw of 18.86 m at the Asian Grand Prix in 2017 is the maximum distance for an Indian female athlete.

Keeping her as a role model, Sanjitha determined to earn gold in Olympics one day. Initially her throw reached 7.56 m only. Being an athlete in school, she regularly practiced both in the mornings and in the evenings and was able to improve the distance by 9 cm every week.

During the special camp for 15 days, she started with 40 throws and every day kept increasing the number of throws by 12 to achieve this remarkable progress.

(a) How many throws Sanjitha practiced on 11th day of the camp?

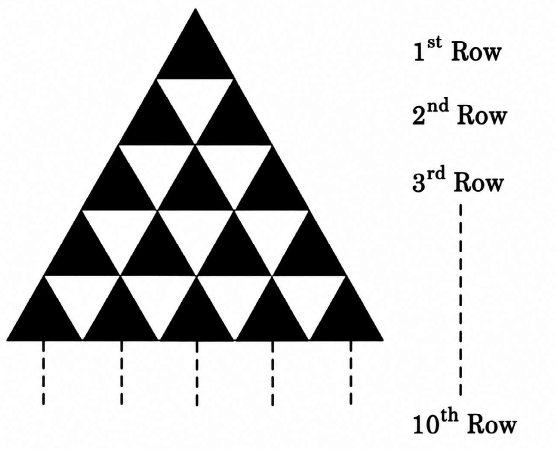
(b) What would be Sanjitha's throw distance at the end of 6 weeks?

OR

When will she be able to achieve a throw of 11.16 m?

(c) How many throws did she do during the entire camp of 15 days? **[CBSE SQP 2023]**

5. In an equilateral triangle of side 10 cm, equilateral triangles of side 1 cm are formed as shown in the figure below, such that there is one triangle in the first row, three triangles in the second row, five triangles in the third row and so on. **[CBSE 2025]**



Based on given information, answer the following questions using Arithmetic Progression.

(i) How many triangles will be there in bottom most row?

(ii) How many triangles will be there in fourth row from the bottom?

(iii) (a) Find the total number of triangles of side 1 cm each till 8^{th} row.

OR

(iii) (b) How many more number of triangles are there from 5^{th} row to 10^{th} row than in first 4 rows ? Show working.

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Answer Key

Exercise-1

Part - 1 (Basic Multiple Choice Questions)

(1) A	(2) B	(3) B	(4) A	(5) A	(6) B
(7) B	(8) B	(9) A	(10) A	(11) A	(12) A

Part - 2 (Basic Subjective Type Questions)

(1) $\frac{3}{2}, \sqrt{2}+\frac{9}{2}, \sqrt{2}+6$

(2) $\frac{5}{2}, 2+\frac{3}{2}, 1$

(3) 25th

(4) $\frac{305}{3}$

(5) a = 2, d =3

(6) 442

(7) 3,4,5,6,...

Exercise-2

Part - 1 (Advanced Basic Multiple Choice Questions)

(1) B	(2) A	(3) B	(4) C	(5) B	(6) B
(7) A	(8) A	(9) B	(10) A	(11) B	(12) A
(13) B	(14) A	(15) A	(16) A	(17) A	(18) A
(19) A	(20) C	(21) A	(22) B	(23) A	(24) D
(25) A	(26) A				

Part - 2 (Advanced Subjective Type Questions)

(1) Proof	(2) 2,6,10,14	(3) Proof	(4) 2,4,6
(5) (i) 550	(ii) 775	(iii) 4375	(6) Proof
(7) n = 16, d = 8/3	(8) 12 terms	(9) –13,-8,-3	
(10) 143 cm	(11) 10 rows, 140 plants	(12) Proof	
(13) Proof	(14) 852	(15) 2,5,8	(16) 100
(17) 25 days	(18) No (n = 9.5)	(19) No	

Part - 3 (Advanced Miscellaneous Questions)

(1) A (2) C (3) D (4) B

(5) A–iii, B–iv, C–i, D–ii (6) A–iii, B–iv, C–i, D–ii

(7) False (8) False (9) True (10) False (11) False

Exercise-3

(Case Based Questions)

(1) (a) A.P. is 3, 6, 9, 12.........., 24, common difference $d = 6 - 3 = 3$

(b) It is not possible to have 34 jars in a layer if the given pattern is continued.

(c) $S_8 = 3 \times \frac{8}{2}[1 + 8] = 108$

(d) $t_5 = 18$

(2) (a) Yes thus average monthly velocity forms an arithmetic progression.

(b) The value of p is 433575 km/hour.

(c) 109125km/hour

(d) 9 months

(3) (a) 120

(b) No. of rows needed to be there = 15

OR

The number of seats still required to be put are 1500 – 750 = 750

(c) 110

(4) (a) 160 throws

(b) Sanjitha's throw distance at the end of 6 weeks = 8.1 m

OR

Sanjitha will be able to throw 11.16m in 41 weeks.

(c) 1860 throws.

(5) (i) There will be 19 triangles in the bottom-most row.

(ii) There will be 13 triangles in the fourth row from the bottom.

(iii) (a) $S_8 = 64$

(iii) (b) Difference = 68

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